

Kunal Chadha

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EDUCATION

Stony Brook University

Master of Science in Computer Science

Stony Brook, NY

Aug 2024 - May 2026 (Expected)

GPA: 4.0, Relevant courses: Computer Architecture, Distributed Systems, Operating Systems, Compilers, Natural Language Processing

Pune Institute of Computer Technology

Bachelor of Engineering in Information Technology

Pune, India

Aug 2017 - May 2021

SKILLS

Domains: Distributed Systems, AI Inference, AI Compilers, GPU Programming, Cloud (AWS, Azure), Natural Language Processing

Programming Languages: C, C++, Python, SystemVerilog, Linux, Ruby

Frameworks & Libraries: PyTorch, CUDA, LLVM & MLIR, Kubernetes, Ansible, Terraform, Docker, Django (Python), Ruby on Rails, PostgreSQL, Elasticsearch, Prometheus, Grafana, Loki

PROFESSIONAL EXPERIENCE

Center Val Bio

Database Development Research Assistant

Stony Brook, NY

Feb 2025 – Present

- Developing a distributed **Django**, **PostgreSQL**, and **Angular** app to manage and search over **500GB+** of **anthropological data**.
- Set up the app's infrastructure on **Azure** and an **on-premise Proxmox cluster** in Madagascar to support local researchers. Used tools like **Ansible**, **Docker Swarm**, **Grafana**, **Loki**, and **GitHub Actions** to automate provisioning, monitoring, and logging.
- Built a replicated **PostgreSQL cluster** using **Patroni** across both clouds to ensure **offline access** for researchers in Madagascar.

BrowserStack Software Pvt. Ltd.

Senior Software Engineer

Mumbai, India

Jul 2021 – Jul 2024

- Led the development of a **central integrations platform** enabling users to seamlessly integrate BrowserStack products with external tools like **Jira**, **Slack**, **GitHub**, and **Jenkins**. Built on Unified APIs, the platform **reduced development time** for new integrations **by 4-6 weeks per tool**, **cut costs by ~80%**, supported **100+ integrations**, and handled **~300,000 requests daily**.
- Designed and developed key **enterprise features**, including **Audit Logging**, **User Management**, **Data Residency**, **Outbound IP Whitelisting**, **SSO**, and **2FA** (adopted by **100,000+ enterprise users**), which increased enterprise adoption by **~20%**
- Strengthened system security and reliability by mitigating spam attacks using **Cloudflare**, implementing monitoring solutions with **Grafana**, **Prometheus**, **Nagios**, and **OpsGenie**, and developing **automated key rotation services**.

PROJECTS

MLISA RISC-V Extension – COMPAS Lab, Stony Brook University

Jan 2025 – Present

- Developing an **LLVM-based compiler backend** for a custom **RISC-V ISA extension (MLISA)** that adds support for AI operations such as **MatMul**, **LayerNorm**, **Gather**, **Softmax**, and others.
- The compiler lowers **PyTorch static graphs** (using the TOSA IR) into **RISC-V assembly with MLISA instructions** for execution on a custom **AI accelerator** developed at the COMPAS Lab, to validate compiler-generated code and measure performance improvements.
- *Technologies:* C, PyTorch, LLVM, MLIR, RISC-V, CUDA

RISC-V CPU Design

- Designed and implemented a **5-stage pipeline CPU** in **SystemVerilog** based on the **RISC-V 64IM ISA**.
- Integrated **branch prediction**, **hazard handling**, and a **cache hierarchy** to optimize performance and functionality.
- Simulates and runs real **C programs** using **Verilator** for testing and validation.
- *Technologies:* C++, SystemVerilog

Intel Pin Tool to enforce Control Flow Integrity

- Developed a Pin-based **binary instrumentation tool** to enforce **backward-edge CFI** by tracking function calls and validating return targets across direct and indirect calls using a shadow stack.
- Enabled multi-threaded support with per-thread shadow stacks to detect control-flow hijacking by logging return address mismatches
- *Technologies:* C++, Intel PIN